HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2012)

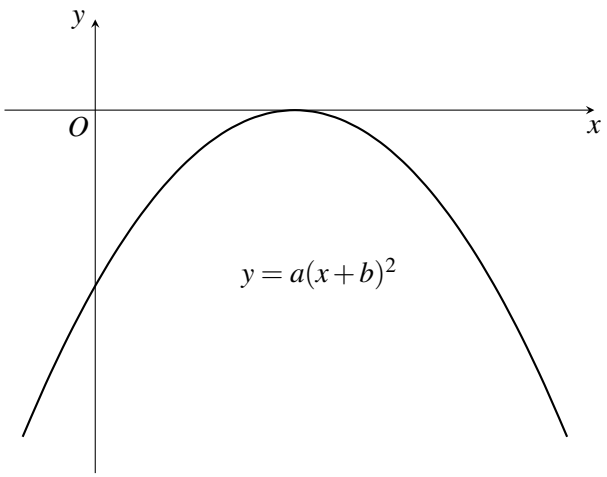
Student No.: 13 Name: _____ Marks: / 119

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The figure shows the graph of $y = a(x+b)^2$, where $a$ and $b$ are constants. Which of the following is true?




A. $a > 0$ and $b > 0$

B. $a > 0$ and $b < 0$

C. $a < 0$ and $b > 0$

D. $a < 0$ and $b < 0$

(DSE-2012 PII Q06)

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2. In a company, 37.5% of the employees are female. If 60% of the male employees and 80% of the female employees are married, then the percentage of married employees in the company is

A. 32.5%.

B. 45%.

C. 55%.

D. 67.5%.

(DSE-2012 PII Q08)

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3. The length of a piece of thin string is measured as 25 m correct to the nearest m. If the string is cut into $n$ pieces such that the length of each piece is measured as 5 cm correct to the nearest cm, find the greatest possible value of $n$.

A. 445

B. 566

C. 567

D. 650

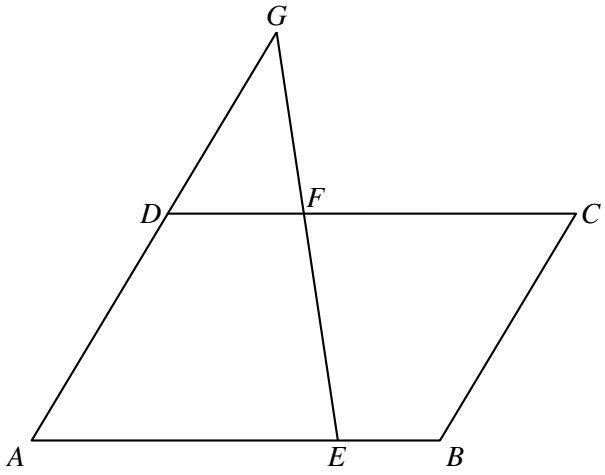
(DSE-2012 PII Q14)

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4. In the figure, $ABCD$ is a parallelogram. $E$ and $F$ are points lying on $AB$ and $CD$ respectively. $AD$ produced and $EF$ produced meet at $G$. It is given that $DF : FC = 3 : 4$ and $AD : DG = 1 : 1$. If the area of $\triangle DFG$ is 3 cm$^2$, then the area of the parallelogram $ABCD$ is




A. 12 cm$^2$.

B. 14 cm$^2$.

C. 18 cm$^2$.

D. 21 cm$^2$.

(DSE-2012 PII Q17)

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5. $\dfrac{\cos 60^\circ}{1-\cos(90^\circ-\theta)} + \dfrac{\cos 240^\circ}{1-\cos(270^\circ-\theta)} =$

A. $\dfrac{1}{\cos^2\theta}$

B. $\dfrac{\cos\theta}{\tan\theta}$

C. $\dfrac{\tan\theta}{\cos\theta}$

D. $\dfrac{1}{\cos\theta\tan\theta}$

(DSE-2012 PII Q19)

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6. Which of the following statements about a regular 12-sided polygon are true?

I. Each exterior angle is 30°.

II. Each interior angle is 150°.

III. The number of axes of reflectional symmetry is 6.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q22)

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7. The rectangular coordinates of the point $P$ are $(-3, -3\sqrt{3})$. If $P$ is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(3, 150°)$.

B. $(3, 330°)$.

C. $(6, 150°)$.

D. $(6, 330°)$.

(DSE-2012 PII Q23)

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8. The stem-and-leaf diagram below shows the distribution of the ages of a group of members in a recreational centre.

| Stem (tens) | Leaf (units) |
|---|---|
| 5 | 0 5 6 6 8 |
| 6 | 1 4 5 5 7 8 8 9 |
| 7 | 3 4 4 6 7 9 |
| 8 | |
| 9 | 1 |

A member is randomly selected from the group. Find the probability that the selected member is not under the age of 74.

A. 0.2

B. 0.3

C. 0.7

D. 0.8

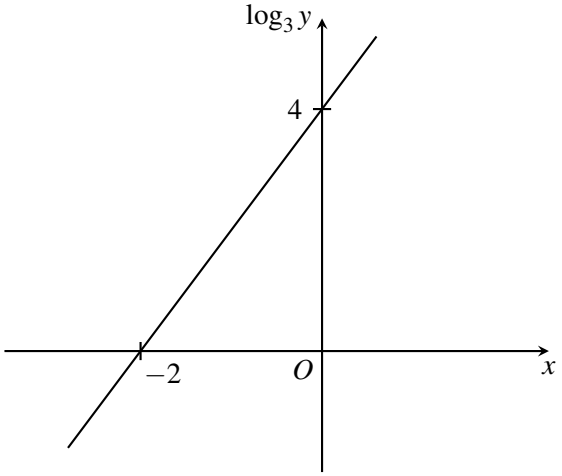
(DSE-2012 PII Q28)

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9. The graph in the figure shows the linear relation between $x$ and $\log_3 y$. If $y = mn^x$, then $n =$




A. $\frac{1}{81}$.

B. $\frac{1}{9}$.

C. 9.

D. 81.

(DSE-2012 PII Q32)

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10. $AD000002012_{16} =$

A. $(10)16^{11} + (13)16^{10} + 8210.$

B. $(10)16^{12} + (13)16^{11} + 131360.$

C. $(11)16^{11} + (14)16^{10} + 8210.$

D. $(11)16^{12} + (14)16^{11} + 131360.$

(DSE-2012 PII Q33)

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11. Let $f(x)$ be a quadratic function. If the coordinates of the vertex of the graph of $y = f(x)$ are $(3, -4)$, which of the following must be true?

A. The roots of the equation $f(x) = 0$ are integers.

B. The roots of the equation $f(x) - 3 = 0$ are rational numbers.

C. The roots of the equation $f(x) + 4 = 0$ are real numbers.

D. The roots of the equation $f(x) + 5 = 0$ are nonreal numbers.

(DSE-2012 PII Q34)

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12. $i^3(3i-3) =$

A. $3+3i$.

B. $3-3i$.

C. $-3+3i$.

D. $-3-3i$.

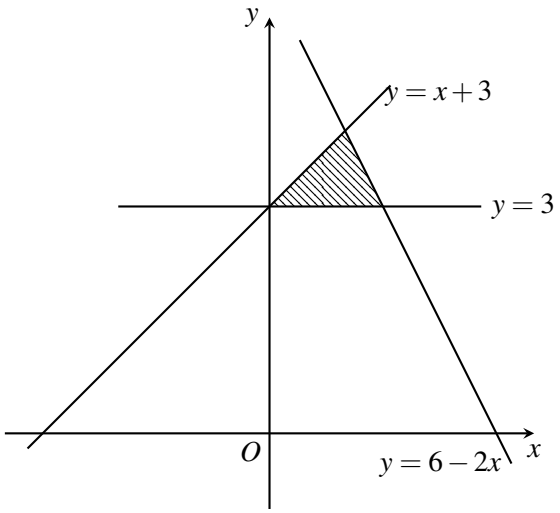
(DSE-2012 PII Q35)

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13. The figure shows a shaded region (including the boundary). If $(h,k)$ is a point lying in the shaded region, which of the following are true?




I. $k \geq 3$

II. $h - k \geq -3$

III. $2h + k \leq 6$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q36)

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14. Let $a_n$ be the $n$th term of an arithmetic sequence. If $a_{18} = 26$ and $a_{23} = 61$, which of the following are true?

I. $a_{14} < 0$

II. $a_1 - a_2 < 0$

III. $a_1 + a_2 + a_3 + \cdots + a_{27} > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q37)

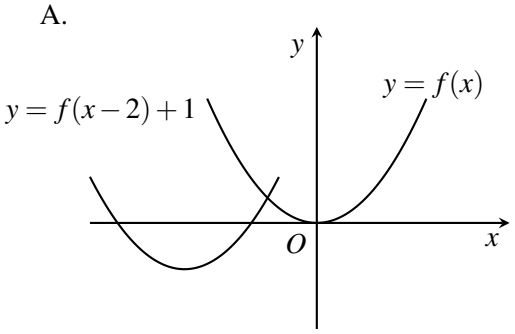
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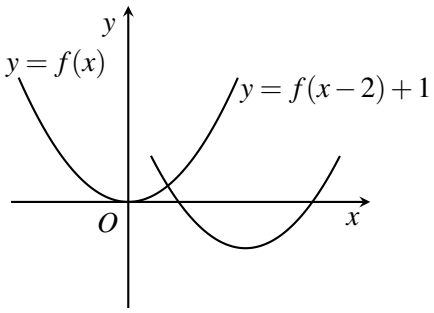
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15. Which of the following may represent the graph of $y = f(x)$ and the graph of $y = f(x-2)+1$ on the same rectangular coordinate system?

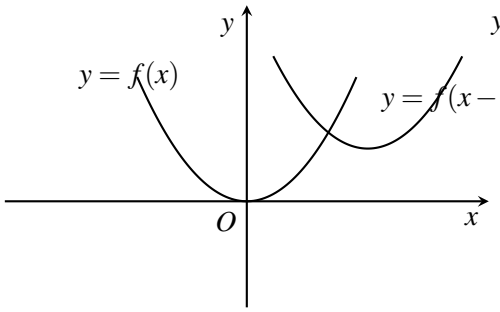
A.



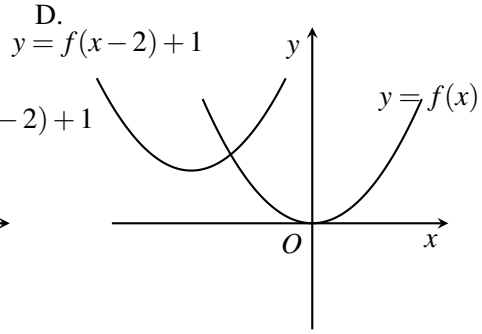

B.




C.




D.



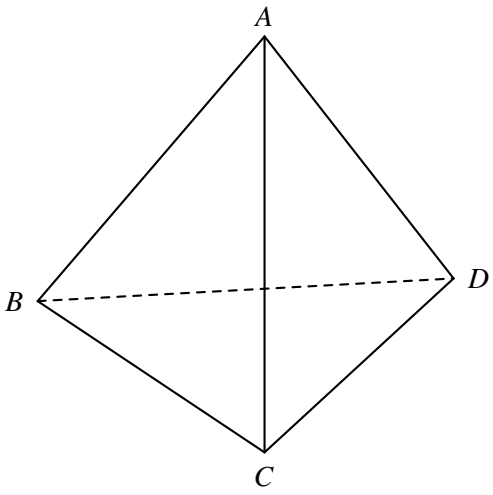

(DSE-2012 PII Q38)

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16. The figure shows a regular tetrahedron *ABCD*. Find the angle between the plane *ABC* and the plane *BCD* correct to the nearest degree.




A. 48°

B. 53°

C. 60°

D. 71°

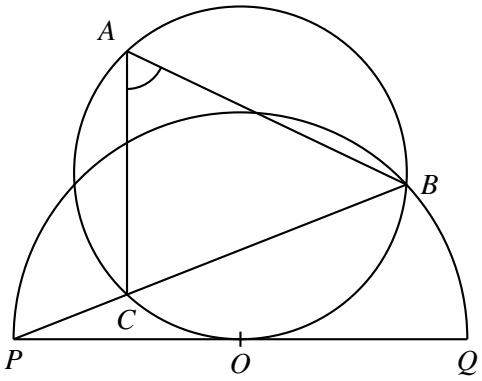
(DSE-2012 PII Q40)

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17. In the figure, $PQ$ is the tangent to the circle $ABC$ at $O$, where $O$ is the centre of the semicircle $PBQ$. It is given that $BCP$ is a straight line. If $\angle BPQ = 12°$, then $\angle BAC =$




A. 18°.

B. 24°.

C. 36°.

D. 54°.

(DSE-2012 PII Q41)

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18. A box contains six balls numbered 7, 8, 8, 9, 9 and 9 respectively. John repeats drawing one ball at a time randomly from the box without replacement until the number drawn is 9. Find the probability that he needs exactly three draws.

A. $\frac{1}{2}$

B. $\frac{1}{6}$

C. $\frac{1}{8}$

D. $\frac{3}{20}$

(DSE-2012 PII Q44)

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19. Let $m_1$, $r_1$ and $v_1$ be the mean, the range and the variance of a group of numbers $\{x_1, x_2, x_3, \ldots, x_{100}\}$ respectively. If $m_2$, $r_2$ and $v_2$ are the mean, the range and the variance of the group of numbers $\{x_1, x_2, x_3, \ldots, x_{100}, m_1\}$ respectively, which of the following must be true?

I. $m_1 = m_2$

II. $r_1 = r_2$

III. $v_1 = v_2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q45)

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20. The daily wage of Ada is 20% higher than that of Billy while the daily wage of Billy is 20% lower than that of Christine. It is given that the daily wage of Billy is $480.

(a) Find the daily wage of Ada.

(b) Who has the highest daily wage? Explain your answer.

(4 marks) (DSE-2012 PI Q04)

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21. (a) Find the range of values of $x$ which satisfy both $\frac{4x+6}{7} > 2(x-3)$ and $2x-10 \leq 0$.

(b) How many positive integers satisfy both the inequalities in (a)?

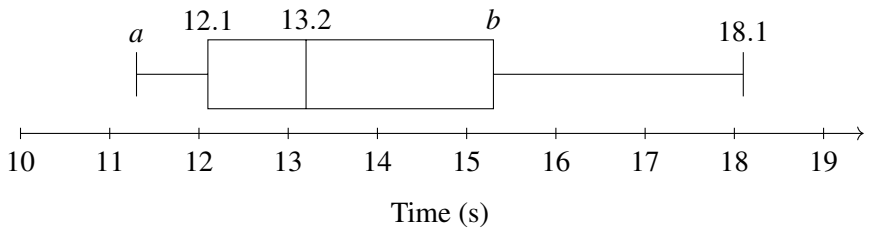
(4 marks) (DSE-2012 PI Q06)

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22. The box-and-whisker diagram below shows the distribution of the times taken by a large group of students of an athletic club to finish a 100 m race:




The inter-quartile range and the range of the distribution are 3.2 s and 6.8 s respectively.

(a) Find $a$ and $b$.

(b) The students join a training program. It is found that the longest time taken by the students to finish a 100 m race after the training is 2.9 s less than that before the training. The trainer claims that at least 25% of the students show improvement in the time taken to finish a 100 m race after the training. Do you agree? Explain your answer.

(4 marks) (DSE-2012 PI Q07)

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23. In Figure 1, $AB$, $BC$, $CD$ and $AD$ are chords of the circle. $AC$ and $BD$ intersect at $E$. It is given that $BE = 8$ cm, $CE = 20$ cm and $DE = 15$ cm.

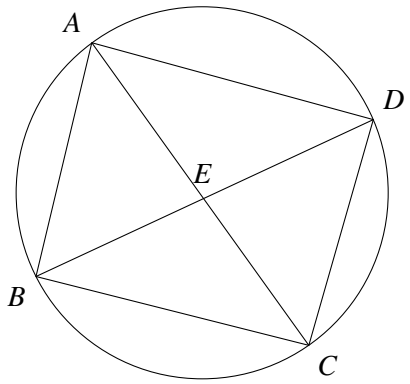



Figure 1

(a) Write down a pair of similar triangles in Figure 1. Also find $AE$.

(b) Suppose that $AB = 10$ cm. Are $AC$ and $BD$ perpendicular to each other? Explain your answer.

(5 marks) (DSE-2012 PI Q08)

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24. In Figure 2, the volume of the solid right prism $ABCDEFGH$ is $1\,020\ \text{cm}^3$. The base $ABCD$ of the prism is a trapezium, where $AD$ is parallel to $BC$. It is given that $\angle BAD = 90°$, $AB = 12$ cm, $BC = 6$ cm and $DE = 10$ cm.

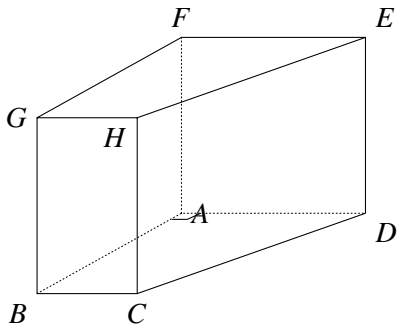



Figure 2

Find

(a) the length of $AD$,

(b) the total surface area of the prism $ABCDEFGH$.

(5 marks) (DSE-2012 PI Q09)

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25. Tom conducts a survey on the numbers of hours spent on doing homework in a week by secondary students. Questionnaires are sent out and twenty of them are returned. The stem-and-leaf diagram below shows the numbers of hours recorded in the twenty questionnaires:

| Stem (tens) | Leaf (units) |
|---|---|
| 1 | 0 0 1 1 2 3 4 5 5 6 6 7 7 |
| 2 | 0 0 0 5 8 |
| 3 | 4 6 |

(a) Find the mean and the median of the numbers of hours recorded in the twenty questionnaires.

(b) Tom receives four more questionnaires. He finds that the mean of the numbers of hours recorded in these four questionnaires is 18. It is found that the numbers of hours recorded in two of these four questionnaires are 19 and 20.

(i) Write down the mean of the numbers of hours recorded in the twenty-four questionnaires.

(ii) Is it possible that the median of the numbers of hours recorded in the twenty-four questionnaires is the same as the median found in (a)? Explain your answer.

(6 marks) (DSE-2012 PI Q10)

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26. Let \$C be the cost of painting a can of surface area $A$ m$^2$. It is given that $C$ is the sum of two parts, one part is a constant and the other part varies as $A$. When $A = 2$, $C = 62$; when $A = 6$, $C = 74$.

(a) Find the cost of painting a can of surface area 13 m$^2$.

(b) There is a larger can which is similar to the can described in (a). If the volume of the larger can is 8 times that of the can described in (a), find the cost of painting the larger can.

(6 marks) (DSE-2012 PI Q11)

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27. (a) Find the value of $k$ such that $x-2$ is a factor of $kx^3-21x^2+24x-4$.

(b) Figure 4 shows the graph of $y=15x^2-63x+72$. $Q$ is a variable point on the graph in the first quadrant. $P$ and $R$ are the feet of the perpendiculars from $Q$ to the $x$-axis and the $y$-axis respectively.

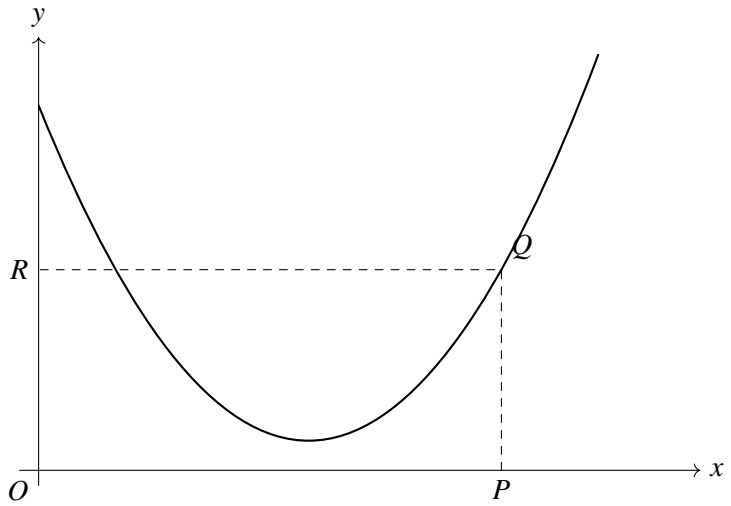



Figure 4

(i) Let $(m,0)$ be the coordinates of $P$. Express the area of the rectangle $OPQR$ in terms of $m$.

(ii) Are there three different positions of $Q$ such that the area of the rectangle $OPQR$ is 12? Explain your answer.

(7 marks) (DSE-2012 PI Q13)

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28. The $y$-intercepts of two parallel lines $L$ and $\ell$ are $-1$ and $-3$ respectively and the $x$-intercept of $L$ is 3. $P$ is a moving point in the rectangular coordinate plane such that the perpendicular distance from $P$ to $L$ is equal to the perpendicular distance from $P$ to $\ell$. Denote the locus of $P$ by $\Gamma$.

(a) (i) Describe the geometric relationship between $\Gamma$ and $L$.

(ii) Find the equation of $\Gamma$.

(b) The equation of the circle $C$ is $(x-6)^2+y^2=4$. Denote the centre of $C$ by $Q$.

(i) Does $\Gamma$ pass through $Q$? Explain your answer.

(ii) If $L$ cuts $C$ at $A$ and $B$ while $\Gamma$ cuts $C$ at $H$ and $K$, find the ratio of the area of $\triangle AQH$ to the area of $\triangle BQK$.

(9 marks) (DSE-2012 PI Q14)

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29. The standard deviation of the test scores obtained by a class of students in a Mathematics test is 10 marks. All the students fail in the test, so the test score of each student is adjusted such that each score is increased by 20% and then extra 5 marks are added.

(a) Find the standard deviation of the test scores after the score adjustment.

(b) Is there any change in the standard score of each student due to the score adjustment? Explain your answer.

(3 marks) (DSE-2012 PI Q15)

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30. The coordinates of the centre of the circle $C$ are $(6, 10)$. It is given that the $x$-axis is a tangent to $C$.

(a) Find the equation of $C$.

(b) The slope and the $y$-intercept of the straight line $L$ is $-1$ and $k$ respectively. If $L$ cuts $C$ at $A$ and $B$, express the coordinates of the mid-point of $AB$ in terms of $k$.

(7 marks) (DSE-2012 PI Q17)

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31. Figure 5(a) shows a right pyramid $VABCD$ with a square base, where $\angle VAB = 72°$. The length of a side of the base is 20 cm. Let $P$ and $Q$ be the points lying on $VA$ and $VD$ respectively such that $PQ$ is parallel to $BC$ and $\angle PBA = 60°$. A geometric model is made by cutting off the pyramid $VPBCQ$ from $VABCD$ as shown in Figure 5(b).

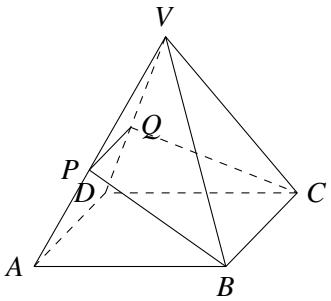



Figure 5(a)

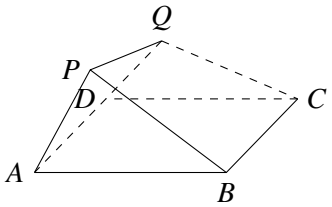



Figure 5(b)

(a) Find the length of $AP$.

(b) Let $\alpha$ be the angle between the plane $PBCQ$ and the base $ABCD$.

(i) Find $\alpha$.

(ii) Let $\beta$ be the angle between $PB$ and the base $ABCD$. Which one of $\alpha$ and $\beta$ is greater? Explain your answer.

(8 marks) (DSE-2012 PI Q18)

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32. In a city, the air cargo terminal $X$ of an airport handles goods of weight $A(n)$ tonnes in the $n$th year since the start of its operation, where $n$ is a positive integer. It is given that $A(n) = ab^{2n}$, where $a$ and $b$ are positive constants. It is found that the weights of the goods handled by $X$ in the 1st year and the 2nd year since the start of its operation are 254 100 tonnes and 307 461 tonnes respectively.

(a) (i) Find $a$ and $b$. Hence find the weight of the goods handled by $X$ in the 4th year since the start of its operation.

(ii) Express, in terms of $n$, the total weight of the goods handled by $X$ in the first $n$ years since the start of its operation.

(b) The air cargo terminal $Y$ starts to operate since $X$ has been operated for 4 years. Let $B(m)$ tonnes be the weight of the goods handled by $Y$ in the $m$th year since the start of its operation, where $m$ is a positive integer. It is given that $B(m) = 2ab^m$.

(i) The manager of the airport claims that after $Y$ has been operated, the weight of the goods handled by $Y$ is less than that handled by $X$ in each year. Do you agree? Explain your answer.

(ii) The supervisor of the airport thinks that when the total weight of the goods handled by $X$ and $Y$ since the start of the operation of $X$ exceeds 20 000 000 tonnes, new facilities should be installed to maintain the efficiency of the air cargo terminals. According to the supervisor, in which year since the start of the operation of $X$ should the new facilities be installed?

(13 marks) (DSE-2012 PI Q19)

I can finish _____ questions

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